

CSMFAB Materials Study — Part VI

In-House Synthesis, Recycled Material Pathways, and Phoenix Protocol Circular Economy

Carrington Storm Motors | Safe Pod Engineering Company | June 2026

Study Scope: This document synthesizes all in-house production opportunities, recycled-material acquisition strategies, and the complete Phoenix Protocol circular economy. It provides the definitive prioritized capital investment roadmap for materials self-sufficiency.

Part A — Complete In-House Production Priority Matrix

The following table consolidates all in-house synthesis opportunities across Parts I–V, ranked by total annual savings potential at full portfolio production scale (5,000-50,000 units/year).

A.1 Tier 1 In-House Programs — Highest ROI

Material	Annual Portfolio Volume	Commercial Cost/kg	In-House Cost/kg	Annual Savings	Capital Required
Basalt fiber	45,000 kg	\$8/kg avg	\$2.10/kg	\$266,000	\$250,000–\$350,000
MXene Ti3C2Tx (from in-house MAX)	120 kg	\$2,000/kg	\$65/kg	\$234,300	Included in MAX Phase
MAX Phase Ti3AlC2	800 kg	\$280/kg avg	\$45/kg	\$188,000	\$35,000 (shared furnace)
KNbO3-BaTiO3 piezo	800 kg	\$200/kg avg	\$35/kg	\$132,000	\$25,000 (shared lab)
ZrB2 powder (SHS)	3,200 kg	\$280/kg	\$102/kg	\$570,000	\$28,000
CoAl2O4 pigment	1,500 kg	\$75/kg	\$30/kg	\$67,500	\$14,000 (shared reactor)
Si3N4 (direct nitridation)	4,500 kg	\$30/kg avg	\$11/kg	\$85,500	\$22,000

Tier 1 combined annual savings at full portfolio: \$1,543,300 Combined capital investment: ~\$374,000 Payback period: 3 months at scale

A.2 Tier 2 In-House Programs — Good ROI

Material	Annual Volume	Commercial \$/kg	In-House \$/kg	Annual Savings	Capital
Aerogel blanket (polyimide-silica)	4,800 m ²	\$250/m ²	\$70/m ²	\$864,000	\$55,000–\$90,000

Material	Annual Volume	Commercial \$/kg	In-House \$/kg	Annual Savings	Capital
CNC aerogel (wood pulp)	2,200 m ²	\$160/m ²	\$35/m ²	\$275,000	\$20,000
3Y-TZP ZrO ₂ (co-precip.)	2,800 kg	\$50/kg avg	\$32/kg	\$50,400	\$20,000
PMMA POF (draw in-house)	50,000 m	\$0.45/m avg	\$0.12/m	\$16,500	\$25,000
AgNW ink	200 kg	\$1,500/kg	\$185/kg	\$263,000	\$18,000
MR Fluid (from BASF CIP)	2,400 kg	\$275/kg	\$32/kg	\$583,200	\$10,000
STF (from Stöber SiO ₂)	600 L	\$45/L	\$15/L	\$18,000	\$8,000
h-BN powder	300 kg	\$38/kg	\$12/kg	\$7,800	\$5,000

Tier 2 combined annual savings: \$2,077,900 Combined capital investment: ~\$201,000 Total Tier 1+2 savings: \$3,621,200/yr at full production scale

A.3 Tier 3 — In-House Assembly and Processing (Not Synthesis)

Process	Annual Volume	External Cost	In-House Cost	Savings	Capital
BFRP panel VARTM	12,000 m ² panels	\$45/m ²	\$12/m ²	\$396,000	\$40,000
BFRP pultrusion (rebar)	500,000 m	\$8/m	\$3/m	\$2,500,000	\$180,000
PEEK machining	8,000 parts	\$45/part avg	\$18/part	\$216,000	\$65,000
Soapstone shaping	4,000 kg	\$55/kg	\$22/kg	\$132,000	\$8,000
POF assembly (sensor ends)	80,000 units	\$2.50/unit	\$0.65/unit	\$148,000	\$12,000
PEN blow molding (spray cans)	500,000 cans	\$0.90/can	\$0.35/can	\$275,000	\$90,000
LiFePO ₄ pack assembly	5,000 packs	\$1.20/Wh	\$0.82/Wh	varies	\$150,000

Part B — The Ideal CSM Manufacturing Facility

B.1 Minimum Viable Production Facility (Phase 1 — \$500K budget)

Facility requirement: Industrial unit, 10,000–15,000 sq ft Location recommendation: Industrial park near University ceramics engineering program (access to shared analytical equipment, talent pipeline)

Essential Equipment for Tier 1 In-House Programs:

Equipment	Function	Cost New	Cost Used
Induction furnace (1600°C, 50L, argon)	ZrB2 SHS, Si3N4, Ti3AlC2	\$85,000	\$25,000
High-energy ball mill (20L)	Powder blending, milling	\$22,000	\$8,000
Spark Plasma Sintering (SPS) press	Densification of ceramics	\$280,000	\$95,000
Hydraulic powder press (50 ton)	Green body formation	\$18,000	\$6,000
Gel-casting/slip casting setup	Complex ceramic shapes	\$15,000	\$5,000
Autoclave reactor (10L, 350°C)	Hydrothermal YInMn, ZrO2	\$28,000	\$12,000
pH-controlled precipitation reactor	3Y-TZP, CoAl2O4 synthesis	\$18,000	\$8,000
Calcination box furnace (1300°C)	Pigment and powder calcination	\$22,000	\$8,000
Ambient-pressure aerogel dryer	Aerogel blanket production	\$15,000	\$6,000
High-shear mixer (100L)	MR fluid, STF formulation	\$12,000	\$4,000
VARTM infusion setup (2 stations)	BFRP panel production	\$35,000	\$12,000
CNC machining center (3-axis)	Soapstone, PEEK, ceramic	\$65,000	\$22,000
XRD (phase verification)	Quality control, ceramics	\$85,000	\$28,000
Laser diffraction (particle size)	Powder characterization	\$45,000	\$15,000
Fume hood + safety equipment	Chemical synthesis	\$25,000	\$10,000

Phase 1 facility equipment total: \$775,000 new / \$264,000 used/refurbished

Phase 1 goes online with: ZrB2 synthesis, Si3N4 synthesis, Ti3AlC2 MAX Phase, CoAl2O4, 3Y-TZP, aerogel, MRF, STF, VARTM BFRP, soapstone machining

B.2 Phase 2 Additions (Year 2–3, \$800K budget)

Equipment	Function	Cost New	Cost Used
Basalt fiber melting furnace (2-ton)	In-house basalt fiber	\$180,000	\$65,000
Pt-Rh fiber drawing bushings	Fiber production	\$80,000	\$35,000
Automated fiber winding system	BFRP rebar pull-winding	\$180,000	\$55,000
MXene synthesis reactor + HF handling	Ti3C2Tx production	\$85,000	\$35,000
Supercritical CO2 dryer (50L)	High-performance aerogels	\$95,000	\$40,000
PEEK injection molding (400°C)	PEEK components	\$75,000	\$28,000

Equipment	Function	Cost New	Cost Used
PEN blow molding (ISBM)	Spray cans	\$90,000	\$35,000
PMMA fiber draw tower	Optical fiber	\$35,000	\$12,000
Roll-to-roll coater (2m wide)	Window film, aerogel	\$380,000	\$120,000
Ionic liquid extraction skid	Phoenix Protocol In/Y	\$65,000	\$28,000
XPS / SEM-EDS analytical	Surface characterization	\$180,000	\$65,000

Phase 2 adds: Full basalt fiber; MXene; advanced aerogel; PEEK molding; window film; Phoenix Protocol full recovery

Phase 2 total: \$1,445,000 new / **\$518,000 used/refurbished**

Part C — Complete Recycled Material Pathway Map

C.1 Primary CSMFAB Materials by Recyclability

Category A — Fully Recoverable (>95% material recovery):

Material	Recovery Method	Recovery Rate	Processing Cost	Recovered Value vs. Virgin
ZrB ₂ -SiC ceramic	Grind □ re-SPS + B ₄ C addition	94% property retention	\$18/kg reprocess	\$85-95/kg vs. \$280/kg virgin
Elium® resin from BFRP	Thermal solvolysis 350°C, 2h	100% monomer	\$8/kg	\$4-6/kg vs. \$8.50-18/kg virgin
Basalt fiber from BFRP	Same thermal solvolysis as Elium	95% fiber yield	\$1.50/kg	\$1.50/kg vs. \$4.50-8/kg virgin
3Y-TZP ZrO ₂ ceramics	HCl dissolution □ re-precipitate	97% Zr yield	\$12/kg	\$18-22/kg vs. \$35-140/kg virgin
MRF-140CG fluid	Drain + centrifugal separation	100%	\$8/L	\$32/kg vs. \$275/kg commercial
KNbO ₃ -BaTiO ₃ piezo	HNO ₃ dissolution □ co-precip	97% K, Nb, Ba, Ti	\$22/kg	\$30/kg vs. \$180-250/kg virgin
LiFePO ₄ cells	Hydrometallurgical □ LFP precur	92% Li, 95% Fe, P	\$14/kg recovd	\$0.70/Wh vs. \$0.85/Wh virgin

Category B — High-Value Critical Mineral Recovery:

Material	Recovery Method	Rate	Key Value
Indium from YInMn glazes	H2SO4 leach □ ionic liquid [P8888][Cl]	99.4%	\$290-380/kg vs. \$18-35/kg recovery cost
Yttrium from YSZ ceramics	HCl dissolution □ Cyanex 272 SX	98%	\$38-55/kg vs. \$22-32/kg recovery cost
Cobalt from CoAl2O4	HCl dissolution □ electroplating	98%	\$32-45/kg Co vs. \$12/kg recovery
Tungsten from WC spheres	Alkali fusion □ APT re-precipitation	88%	\$34/kg W vs. \$15/kg recovery
Samarium from SmCo magnets	H2SO4 dissolution □ selective precipitation	92%	\$60-90/kg Sm vs. \$20/kg recovery
Niobium from KNbO3	HNO3 digest □ ammonium niobate precipitation	88%	\$40/kg Nb vs. \$18/kg recovery

Category C — Bio-Based / Agricultural Recyclable:

Material	Source	Recovery	Notes
CNC aerogel (cellulose)	Wood pulp, ag waste	Fully biodegradable	Carbon-negative feedstock
Bio-Elum® (future)	Sugarcane-derived MMA	100% chemical recycle	Braskem Green PE path
Soapstone shavings	CNC dust from machining	100% reused	Reconstitute in CBPC mix

Part D — Phoenix Protocol Master Flow Chart

D.1 Collection and Disassembly

Collection triggers: - End-of-service product return (CSM take-back program — 5-year warranty with recovery incentive) - Component swap for upgrade (Aegis Embark module replacement) - Infrastructure decommissioning (power line poles, pipeline sections)

Disassembly sequence: 1. Electronic/optical components extracted □ electronics refurbishment or component recovery 2. PEEK/ZrO2 hardware removed (ultrasonic cleaning + inspection □ re-use if intact) 3. BFRP/Elum® structural panels □ thermal solvolysis reactor (350°C, 2h) 4. Ceramic cores (ZrB2-SiC, ZrO2) □ crushing and milling / hydrometallurgy 5. YInMn/CoAl2O4 glazes □ acid leach □ critical mineral SX extraction 6. MRF fluid □ drain and centrifuge □ quality certification □ re-fill 7. LiFePO4 battery □ discharge □ hydromet processing

D.2 Phoenix Protocol Economics at Scale

Scenario: 10,000 Aegis-C wall panels (1m² each) retired

Material Stream	Mass	Recovery Rate	Recovered Value
ZrB2-SiC panel core	46,000 kg	94% property retention	\$46,000 × 0.94 × (\$280-\$65) = \$929,880
Elum® monomer	18,000 kg	100%	\$18,000 × \$5/kg = \$90,000

Material Stream	Mass	Recovery Rate	Recovered Value
Basalt fiber	8,000 kg	95%	$8,000 \times \$4.50/\text{kg} = \$36,000$
Indium from YInMn	65 kg	99.4%	$65 \times 0.994 \times (\$320 - \$28/\text{kg recovery}) = \$18,943$
Yttrium from YInMn	41 kg	98%	$41 \times 0.98 \times (\$47 - \$27) = \$803$
Si ₃ N ₄ buffer	76,000 kg	97%	$\$76,000 \times 0.97 \times (\$28 - \$8) = \$1,474,400$
CoAl ₂ O ₄ interior	13,000 kg	98%	$\$13,000 \times 0.98 \times (\$70 - \$15) = \$701,000$
TOTAL RECOVERED			\$3,251,026

vs. Demolition/landfill value of steel-reinforced concrete equivalent: ~\$0 (disposal cost)

Net recovery advantage: \$3.25M per 10,000 panels (vs. standard concrete: \$0)

Part E — Materials for Specific In-House Production — Step-by-Step Guides

E.1 ZrB₂ Powder via SHS — Detailed Protocol

Batch size: 5 kg of ZrB₂ powder per run

Materials: - Zirconium metal powder (99.5%, -100 mesh): 4.28 kg □ Source: APC Pure Chemicals, \$45/kg
- Crystalline boron powder (95%, -325 mesh): 1.30 kg □ Source: Noah Technologies, \$30/kg - Argon gas (industrial, 99.99%): 50L cylinder □ \$35/cylinder - Graphite susceptor (existing): shared equipment

Procedure: 1. Weigh Zr:B in 1:2.05 molar ratio (5% B excess to compensate for B evaporation) 2. Blend in ball mill under argon: 4 hours, 300 rpm 3. Press blended powder into cylindrical pellets at 100 MPa (hydraulic press) 4. Load pellets into argon-purged furnace boat (graphite-lined alumina crucible) 5. Purge furnace 3× with argon 6. Heat to 700°C at 10°C/min — hold 10 min (purge volatile surface oxides) 7. Heat to 1000°C at 15°C/min — SHS ignition typically occurs at 950-1050°C 8. Monitor: exotherm reaches 2200°C for ~30 seconds □ reaction completes 9. Cool under argon to <200°C; no cooling rate restriction (product is stable) 10. Remove product; mill to target particle size (3-5 μm for SPS sintering) 11. XRD verification: confirm ZrB₂ phase (no residual ZrO₂ or B peaks)

Cost per batch: - Zr powder: $\$45/\text{kg} \times 4.28 = \192.60 - B powder: $\$30/\text{kg} \times 1.30 = \39.00 - Argon: \$35 - Milling media wear: \$8 - Energy (furnace, 3h): \$12 - Labor (4h @ \$25/hr): \$100 - **Total: \$386.60 per 5 kg = \$77.32/kg — vs. \$280+ commercial**

E.2 CoAl₂O₄ — Detailed Protocol

Batch size: 2 kg CoAl₂O₄

Materials: - Co(NO₃)₂·6H₂O: 1,740 g □ \$18/kg; needed 3.13 kg □ \$56.34 - Al(NO₃)₃·9H₂O: 6,530 g □ \$4/kg; needed 6.53 kg □ \$26.12 - NH₄OH (28%): 2L □ \$3.50/L □ \$7.00 - Deionized water: 20L (in-house RO) - NaOH for pH adjustment: 50g □ \$0.10

Procedure: 1. Dissolve Co(NO₃)₂ and Al(NO₃)₃ in 10L DI water (molar ratio Co:Al = 1:2) 2. Prepare NH₄OH solution (dilute 1:2 with DI water) in separate vessel 3. Add Co-Al nitrate solution dropwise to NH₄OH with stirring □ blue-green precipitate 4. Maintain pH 9–10 throughout addition 5. Filter precipitate; wash 3× with DI water; filter 6. Dry at 120°C overnight 7. Calcine at 900°C for 4h in air (ceramic furnace — shared) 8. Cool; grind in mortar to pass 100-mesh sieve 9. Color check: deep royal blue (indicates correct CoAl₂O₄ phase) 10. XRD confirmation if batching critical product

Cost per batch: - Co(NO₃)₂: \$56.34 - Al(NO₃)₃: \$26.12 - NH₄OH: \$7.00 - Energy, DI water, consumables: \$12 - Labor (6h at \$25): \$150 - **Total: \$251.46 per 2 kg = \$125.73/kg raw cost + scale-up savings** At 50-kg batches: ~\$35/kg (vs. \$55-100 commercial)

E.3 Silica Aerogel Blanket — Ambient Pressure Route

Target: 1 m² × 10mm blanket, λ = 0.015 W/m·K

Materials per m²: - TEOS (tetraethyl orthosilicate): 380 mL □ \$3.50/100mL □ \$13.30 - Ethanol (95%): 1.2 L □ \$1.20/L □ \$1.44 - HCl (catalyst, 0.1M): 25 mL □ \$0.15 - NH₄OH (base catalyst, 5%): 25 mL □ \$0.20 - MTMS (hydrophobizing agent): 35 mL □ \$8/100mL □ \$2.80 - HMDSO (surface treatment, optional): 15 mL □ \$6/100mL □ \$0.90

Procedure (simplified ambient-pressure route): 1. Prepare base TEOS sol: TEOS + ethanol + HCl + DI water □ stir 30 min 2. Add NH₄OH □ gelation occurs in 5-20 min; pour into flat mold (1m × 1m × 12mm) 3. Age gel 24h in sealed mold at 60°C 4. Solvent exchange: replace pore water with ethanol (3×), then hexane (1×) 5. Add MTMS in hexane □ surface modification 2h (prevents capillary collapse) 6. Evaporate hexane slowly at 60°C for 12h 7. Final drying at 150°C for 2h 8. Result: flexible, hydrophobic aerogel blanket

Cost per m² batch: - TEOS: \$13.30 - Ethanol: \$1.44 - MTMS: \$2.80 - NH₄OH, HCl, other: \$1.20 - HMDSO: \$0.90 - Energy (oven): \$2.50 - Labor (8h @\$25): \$200 (first run; drops to \$45 with experience) - Mold (amort.): \$5 - **First-run cost: ~\$227/m²; steady-state: \$72/m²** (vs. \$120-280/m² commercial)

Part F — Prioritized Capital Investment Roadmap

Year 1: \$264,000 in used equipment (Tier 1 in-house programs)

Priority 1 investment: ZrB₂ SHS program (\$28,000) - SHS reaction vessel, ball mill, basic safety enclosure - Payback from ZrB₂ savings: **5 months**

Priority 2 investment: CoAl₂O₄ + 3Y-TZP synthesis (\$34,000) - Shared reaction vessel, pH electrode, calcination furnace - Annual savings: \$117,900 - Payback: **3.5 months**

Priority 3 investment: Shared tube furnace for Si₃N₄, MAX Phase, KNbO₃-BaTiO₃ (\$22,000) - Covers three high-savings programs on single equipment - Annual savings: \$405,500 - Payback: **3 weeks**

Priority 4 investment: Aerogel blanket line (\$55,000) - Ambient-pressure drying in existing ovens initially - Annual savings: \$864,000 - Payback: **3 weeks**

Priority 5 investment: VARTM BFRP panel station (\$40,000) - Immediate use for all structural BFRP products - Annual savings: \$396,000 - Payback: **1 month**

Year 2: \$518,000 (Phase 2 — basalt fiber, MXene, advanced aerogel)

Highest-priority Year 2 investment: Basalt fiber furnace (\$245,000 total) - Recovers investment in 14-18 months from fiber savings alone - Once operational, BFRP production cost drops dramatically for all composite products

Year 3: Phoenix Protocol Recovery Facility (\$180,000)

- Ionic liquid In/Y extraction
- Thermal solvolysis for Elium®/basalt fiber recovery
- LiFePO₄ hydromet processing
- **Recovery stream generates \$3.25M per 10,000 panel equivalents retired**

Summary: Materials Self-Sufficiency Timeline

Year	Annual In-House Production	Savings vs. Full Buy	Cumulative Investment
Year 1	ZrB2, Si3N4, CoAl2O4, 3Y-TZP, aerogel (AP), BFRP panels	\$1.8M/yr	\$264,000
Year 2	+ Basalt fiber, MXene, MAX Phase, PVDF film, AgNW	\$3.6M/yr	\$782,000
Year 3	+ Full Phoenix Protocol recovery; PEN cans; POF fiber	\$5.2M/yr	\$962,000
Year 4	+ Roll-to-roll window film, advanced aerogel, full ceramic	\$7.1M/yr	\$1.8M
Year 5	Full portfolio self-sufficiency (bought: LoRa, GaN, LiFePO4 cells)	\$8.5M/yr	\$2.4M

By Year 5, CSM produces ~85% of materials in-house or from recycled streams. Materials cost advantage vs. pure-buy portfolio: 55-70% lower COGS. This transforms gross margins from 35-45% (projected) to 55-68% — competitive moat.

End of CSMFAB Materials Study Part VI | In-House Synthesis, Recycled Pathways, Phoenix Protocol Car-rington Storm Motors | Safe Pod Engineering Company | June 2026 All cost estimates based on 2026 market data and standard industrial chemistry process economics.